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Purposefully different,

consistently excellent

Welcome to EEN1037: Web Applications Development

CSS Frameworks: Bootstrap

Week 2

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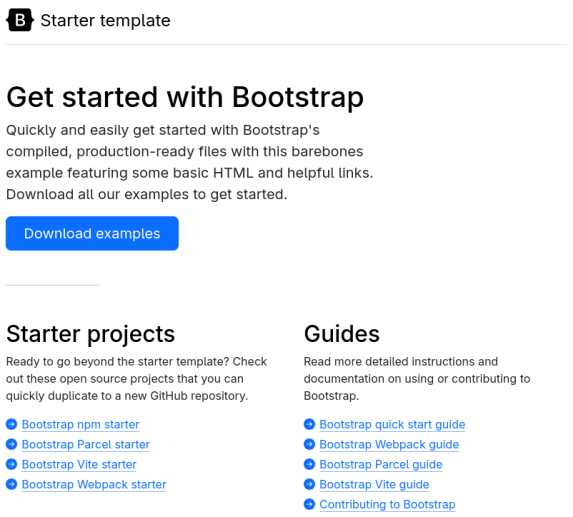
What is Bootstrap?

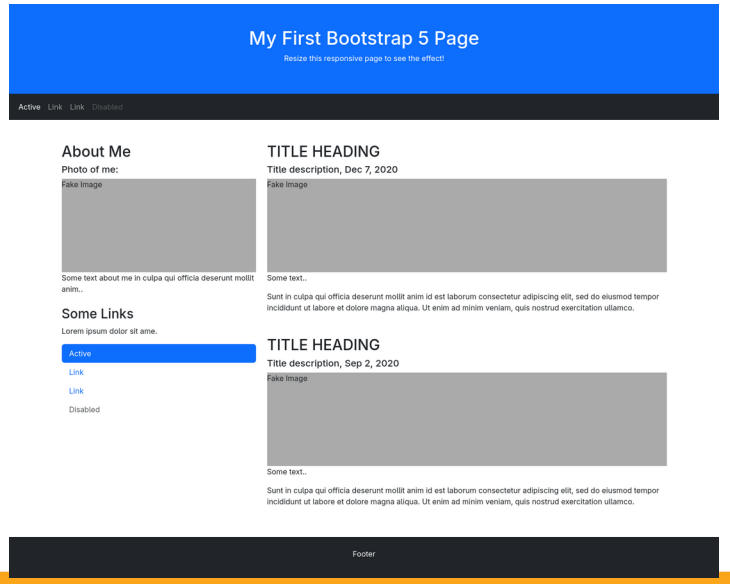
- Ready-to-go CSS framework: layout grid, components, utilities
- Minimal custom CSS for nice & consistent UI look
- Simply add Bootstrap CSS class selectors to HTML element class="" attributes to apply Bootstrap styling and layout. (Can still include your own CSS)
- Saves time and frustration fighting with CSS layout and styling
- Downside is "generic" Bootstrap appearance

```
<!doctype html>
<html lang="en">
<head>
  <title>My Example Page</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
  <div class="container py-4">
    <h1 class="h3 mb-3">Bootstrap Example</h1>
    <p class="text-muted mb-0">Basic bootstrap example.</p>
  </div>
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Getting started with Bootstrap

- Follow the quick-start guide
  - https://getbootstrap.com/docs/5.3/getting-started/introduction/
- Copy the starter template, and adjust to match your needs
  - https://getbootstrap.com/docs/5.3/examples/starter-template/
  - https://www.w3schools.com/bootstrap5/bootstrap_templates.php

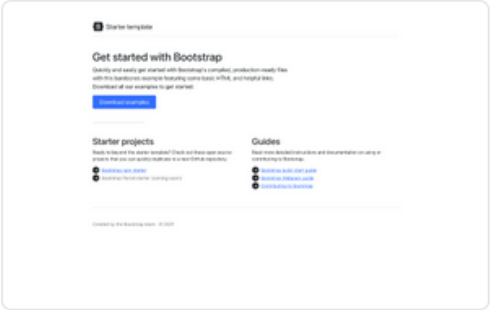




Examples & Reference Documentation

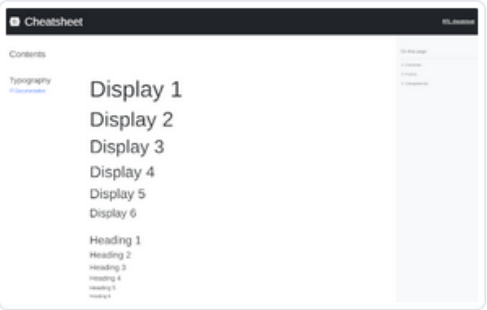
- Project documentation is very good:
  - https://getbootstrap.com/docs/5.3
- Visual examples are the best way to find what your looking for:
  - https://getbootstrap.com/docs/5.3/examples/
  - https://getbootstrap.com/docs/5.3/examples/grid/
  - https://getbootstrap.com/docs/5.3/examples/cheatsheet/

Examples that focus on implementing uses of built-in components provided by Bootstrap.



Starter template

Grid



Cheatsheet

Containers & spacing

- Layout wrapper: container, container-fluid
- Spacing utilities: p-*, m-*, py-*, gx-* etc. (0–5)

```
<div class="container p-4">
  <div class="p-3 mb-3 bg-light border rounded">Box A</div>
  <div class="p-3 bg-light border rounded">Box B</div>
</div>
```

Box A

Box B

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Grid basics

- 12-column grid via .row + .col-*
- Distribute evenly: .col

```
<div class="container">
  <div class="row g-3">
    <div class="col-12 col-md-8"><div class="p-3 bg-light border">Main</div></div>
    <div class="col-12 col-md-4"><div class="p-3 bg-light border">Sidebar</div></div>
  </div>
</div>
```

Main (col-12 col-md-8)

Sidebar (col-12 col-md-4)

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Typography

- Headings: .h1–.h6, display: .display-1–.display-6
- Text utilities: .fw-bold, .text-muted, .lh-*

```
<h2 class="display-6 fw-semibold">Title</h2>
<p class="text-muted mb-0">Muted body text.</p>
```

Title

Muted body text.

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Colors & Backgrounds

- Text: .text-primary, .text-danger, …
- Background: .bg-primary, .bg-dark, …
- Contrast helper: .text-white

```
<div class="p-3 bg-primary text-white rounded">Primary banner</div>
<div class="p-3 bg-warning rounded mt-2">Warning</div>
```

Primary banner

Warning

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Colors & Backgrounds

- .btn + variant (primary, outline-*, danger, …)
- Sizes: .btn-sm, .btn-lg

```
<button class="btn btn-primary">Save</button>
<button class="btn btn-outline-secondary">Cancel</button>
<button class="btn btn-danger btn-sm">Delete</button>
```

Save Cancel Delete

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Forms

- .form-label, .form-control, .form-select
- Layout with grid utilities

```
<form>
  <label class="form-label" for="email">Email</label>
  <input id="email" class="form-control" type="email" placeholder="you@site.com">
  <div class="mt-3">
    <label class="form-label">Plan</label>
    <select class="form-select">
      <option>Free</option><option>Pro</option>
    </select>
  </div>
  <button class="btn btn-primary mt-3">
    Continue
  </button>
</form>
```

Email

you@site.com

Plan

Free

Continue

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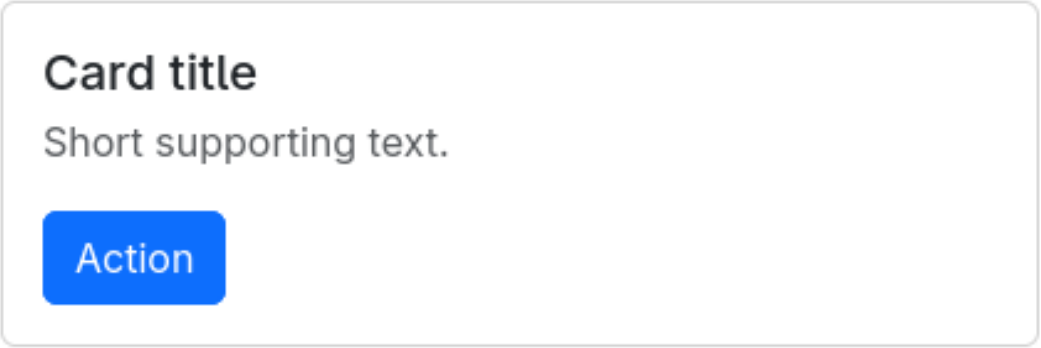
Cards

- Common content container: .card, .card-body, .card-title

```
<div class="card" style="max-width: 400px;">
  <div class="card-body">
    <h5 class="card-title mb-1">Card title</h5>
    <p class="card-text text-muted">Short supporting text.</p>
    <a class="btn btn-primary" href="#">Action</a>
  </div>
</div>
```



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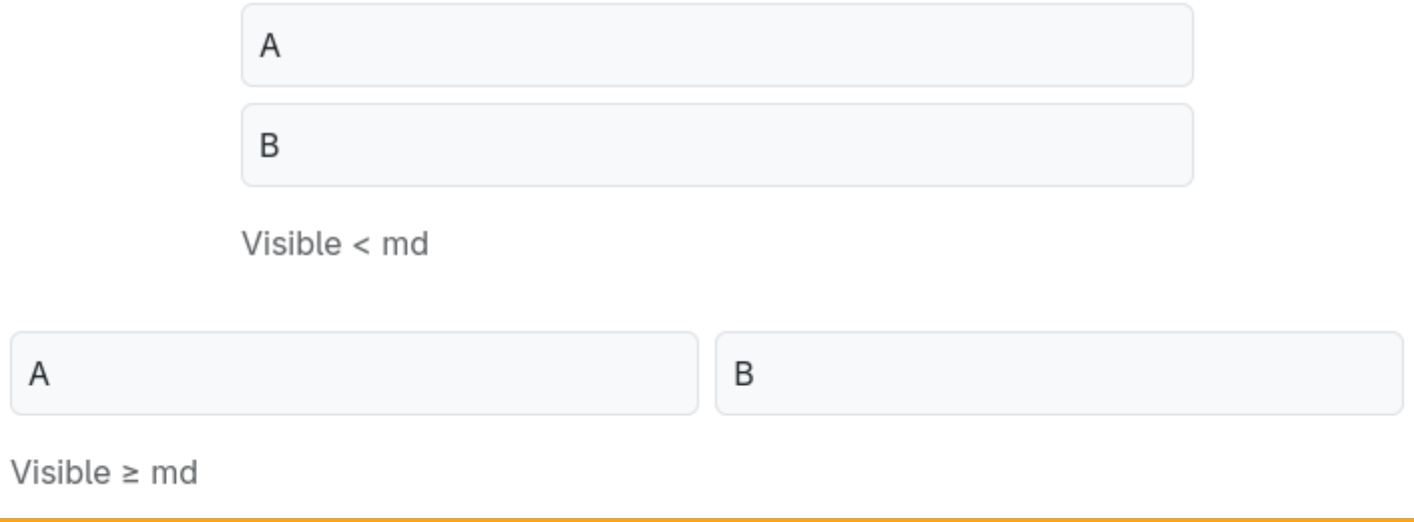
Responsive Utilities

- Display: d-none d-md-block, d-grid
- Flex: d-flex, justify-content-*, align-items-*

```
<div class="d-flex flex-column flex-md-row gap-2">
  <div class="p-2 bg-light border flex-fill">A</div>
  <div class="p-2 bg-light border flex-fill">B</div>
</div>
<p class="d-none d-md-block text-muted mt-3 mb-0">Visible ≥ md</p>
<p class="d-md-none text-muted mt-3 mb-0">Visible &lt; md</p>
```



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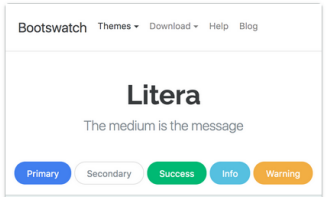
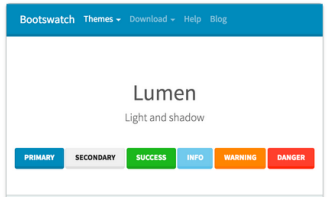
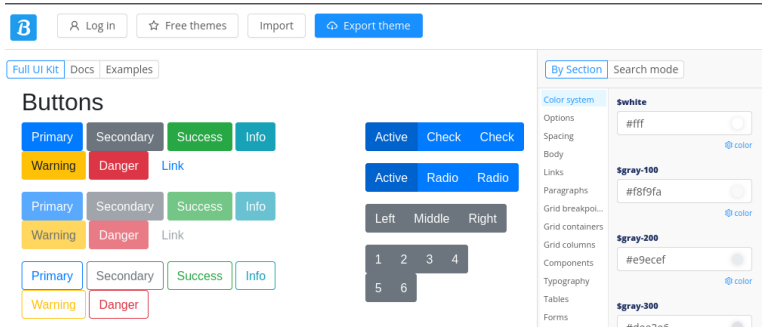
Customizing Bootstrap

- Downside is all Bootstrap sites look the same. OK for internal tools, may not be OK for customer facing sites. How can we change theme appearance, colors etc?
- Bootstrap CSS file is machine generated and complex, hand editing not recommended, leads to unmaintainable mess. [don't do it!]
- Can modify Bootstrap SASS source and recompile [complex]:
  - https://getbootstrap.com/docs/5.3/customize/overview/
- Or use preexisting templates [easy]:
  - https://bootswatch.com/
  - https://cdnjs.com/libraries/bootswatch





- Or create your own theme with an online tool [easy]:
  - https://bootstrap.build/app
  - (Be sure to save _variables.scss and custom.scss)



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